The Association Between Moderate Physical Activity and Self-Reported Depression Outcomes

Evidence from the National Health Interview Survey, 2010–2023

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BACKGROUND

21M
U.S. adults had a major depressive episode in 2021 (8.3%)[1]

280M
people live with depression worldwide[2]

17.2%
of U.S. adults aged 18–25 had depression in 2020, up from 10.3% in 2015[3]

Treatment barriers
- High costs
- Long waitlists
- Provider shortages

→

Physical activity
- Low-cost
- No prescription
- Flexible, anywhere

Prior evidence: activity is linked to lower depression risk

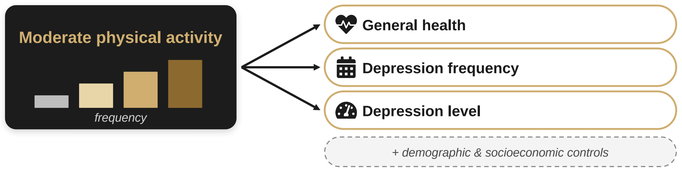
- Any activity[4] — 17% lower odds
- Half the recommended dose[5] — 18% lower risk
- Full recommended dose[5] — 25% lower risk

The gap: few nationally representative studies compare how often people exercise across several dimensions of depression.

AIMS

Question: Is how often U.S. adults do moderate physical activity linked to their self-reported depression and general health?

Hypothesis: Since activity can lower stress and aid sleep, more frequent activity is linked to better depression outcomes.



METHODS

Data: IPUMS NHIS[6], 2010–2023

N = 332,521 U.S. adults · self-reported · nationally representative

Exposure: moderate activity, times/week

None 0 | Low 1–3 | Medium 4–6 | High 7+

Outcomes: 3 self-reported measures

General health | Dep. frequency | Dep. level

Model: OLS regression vs. no activity

(3) + Socioeconomic: marital status, income, poverty ratio, meds ★ *preferred*
(2) + Demographic: age, sex, race
(1) Activity only

How to read: each estimate is the percentage-point difference vs. inactive adults.

RESULTS

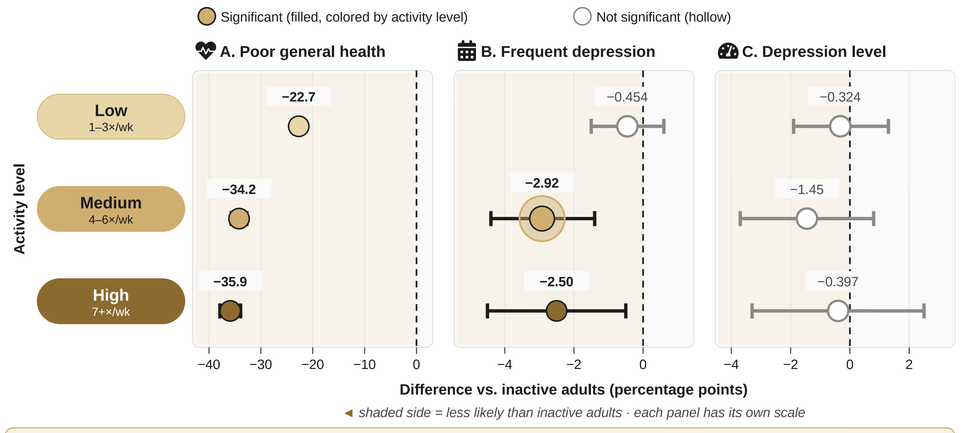
−2.92
percentage points

KEY FINDING

Adults active 4–6 times a week were 2.92 percentage points less likely to report frequent depressive symptoms than inactive adults.

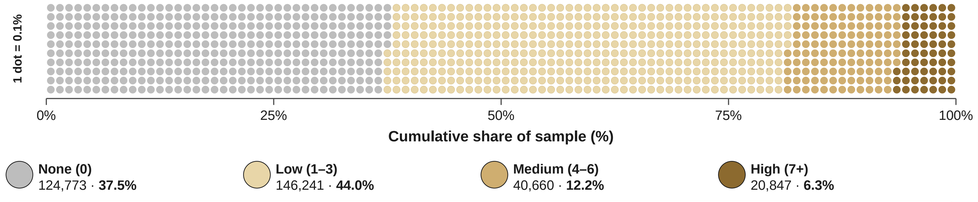
95% CI: −4.4 to −1.4 · Specification 3

Figure 1. Change in each outcome vs. no activity, by activity level (Spec 3, 95% CI)



Takeaway: better general health at every activity level; fewer frequent depressive symptoms only at 4–6 and 7+ sessions a week.

Figure 2. Who was studied: sample by activity level (N = 332,521)



Takeaway: most adults were inactive or low-activity; only 12.2% reached the 4–6 times-a-week level linked to fewer depressive symptoms.

Table 1. Specification 3 estimates, percentage points [95% CI]

Activity level	General health (poor)	Depression frequency	Depression level
Low (1–3)	**−22.7*** [−23.8, −21.6]	−0.454 [−1.5, 0.6]	−0.324 [−1.9, 1.3]
Medium (4–6)	**−34.2*** [−35.7, −32.7]	**−2.92*** [−4.4, −1.4]	−1.45 [−3.7, 0.8]
High (7+)	**−35.9*** [−37.9, −33.9]	**−2.50*** [−4.5, −0.5]	−0.397 [−3.3, 2.5]
Observations	161,344	160,974	66,089
R^2	0.204	0.252	0.038

** Significant (95% CI excludes 0); highlighted. Reference group: no moderate activity.*

DISCUSSION

- **4–6 sessions a week** had the largest link with fewer frequent depressive symptoms
- **Better self-rated health** at every activity level (−22.7 to −35.9 pp)
- **Fits prior work** linking activity to lower depression risk[4,5]
- **Socioeconomic status matters:** welfare receipt consistently predicted worse outcomes[7]
- **No link with depression level:** all CIs include 0 (R^2 = 0.038)

Limitations

- Self-report bias
- Unmeasured confounders
- Association, not cause
- Income only in categories

CONCLUSIONS

1. **Being moderately active 4–6 times a week was linked to a lower chance of feeling depressed often.**
2. Any regular moderate activity – even 1–3 times a week – was linked to better self-rated health.
3. Activity was not linked to how strongly depression is felt, so it is a low-cost complement to care, not a cure.

Big picture: as depression rises and care stays hard to reach, regular moderate activity is a low-cost tool worth promoting.

Future directions
Test activity as a treatment for people clinically diagnosed with depression, tracking real-time, individual-level change.

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Author Contribution Statement
NG (sole author) conceived and designed the study, prepared the data, performed all analyses, interpreted the results, wrote the manuscript, and designed this poster.

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